Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE NUMBER: MT 360A

COURSE TITLE: Calculus I

Course Description Treats standard topics of single variable calculus including limits, continuity, derivatives, applications of derivatives, and elements of integration.

Course Prerequisites MT 201, College Algebra

Course Outcomes

Upon successful completion of this course, students should be able to:

- Demonstrate that the student is able to communicate mathematical concepts so that they may provide the public with information and expert judgment as part of the basis for social and political decisions.
- Demonstrate an awareness of the socially beneficial uses of mathematics, as well as harmful uses of mathematics.
- Define a function and recognize basic functions such as polynomials, trigonometric and inverse trigonometric functions, exponentials, and logarithmic functions.
- Define limits and evaluate limits numerically, graphically, and algebraically.
- Apply the definition of continuity to determine the intervals when a function is continuous.
- Define a tangent line and demonstrate methods for finding its slope and equation.
- Apply direct and indirect methods of differentiation to develop derivatives formulas for logarithmic, exponential, and inverse trigonometric functions.
- Apply derivatives to rate of change problems and use linear functions to approximate nonlinear functions.
- Demonstrate the ability to use methods of calculus to analyze functions and their graphs.
- Apply the Fundamental Theorem of Calculus to calculate the area under a curve.
- Define the average value of a function.

Course Materials and Resources, including Technology Tools

Required:

- Hass, J., Weir, M. D., & Thomas, G. B. (2012). University calculus: Early transcendentals (2nd ed.). Upper Saddle River, NJ: Pearson Education. ISBN13: 978-0-321-75990-0 or ISBN10: 0-321-75990-7 hardcover (book with MyMathLab).
- TI-8x or equivalent graphing calculator.

Method of Instruction

This course will use MyMathLab lab exercises and facilitated discussions.

Related Standards

Related Standards	
	This program enables students to achieve the following program outcomes by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others

First Week Assignment

1. Read Chapter 1 in the text.
2. Write a two-page, double-spaced, paper on the topic of your math background. Topics that should be included are what math courses you have taken in the past, what your major is, and how this course will fit into that major. The end result must be a well-organized and written document of the specified length. Anything less may result in having to rewrite the paper.

Course Overview Grid

Week	Learning Topics	Activities	Reading Assignments	Assignments
1	Functions • Functions and Graphs • Combining Functions • Trigonometric Functions • Exponential Functions • Inverse Functions and Logarithms • Graphing with Calculators and Computers	Discussion Forum: Topic 1 and 2 Lab Exercises (MyMathLab)	Chapter 1	First week assignment MyMathLab exercises (solve as assigned) Week 1 Quiz
2	Limits and Continuity • Rates of Change • Limit of a Function • Limit Laws • Definition of a Limit	Discussion Forum: Topic 1 Lab Exercises (MyMathLab)	Chapter 2, Sections 1-3	MyMathLab exercises (solve as assigned) Week 2 Quiz
3	Limits and Continuity • One-Sided Limits • Continuity • Tangents and Derivatives	Discussion Forum: Topic 1 Lab Exercises (MyMathLab)	Sections 2.4-2.6 and Section 3.1	MyMathLab exercises (solve as assigned) Week 3 Quiz
4	Derivatives • Derivative as a Function • Differentiation Rules • Derivative as a Rate of Change • Derivatives of Trigonometric Functions • Chain Rule and Parametric Equations	Discussion Forum: Topic 1, 2, and 3 Lab Exercises (MyMathLab)	Sections 3.2-3.6	MyMathLab exercises (solve as assigned) Week 4 Quiz
5	Derivatives • Implicit Differentiation • Derivatives of Inverse Functions and Logarithms	Discussion Forum: Topic 1 and 2 Lab Exercises	Sections 3.7-3.11	MyMathLab exercises (solve as assigned) Week 5 Quiz

Week	Learning Topics	Activities	Reading Assignments	Assignments
	• Derivatives of Logarithms and Exponential Functions • Derivatives of Inverse Trigonometric Functions. • Related Rules • Linearization and Differentials	(MyMathLab)		
6	Applications of Derivatives • Minimum and Maximum Values of Functions • Critical Points • The Mean Value Theorem • The First Derivative Test • Concavity	Discussion Forum: Topic 1 Lab Exercises (MyMathLab)	Sections 4.1-4.4	MyMathLab exercises (solve as assigned) Week 6 Quiz
7	Applications of Derivatives • Applied Optimization • Indeterminate Forms and L'Hopital's Rule • Newton's Method • Antiderivatives	Discussion Forum: Topic 1 Lab Exercises (MyMathLab)	Sections 4.5-4.8	MyMathLab exercises (solve as assigned) Week 7 Quiz
8	Introduction to Integration • Estimating with Finite Sums • Sigma Notation, Rieman Sums and the Definite Integral • Fundamental Theorem of Calculus	Discussion Forum: Topic 1 Lab Exercises (MyMathLab)	Sections 5.1-5.4	MyMathLab exercises (solve as assigned) Final Exam

Student Evaluation Grid

Assignment	Value (percent of overall course grade)
Weekly Homework for Grade(eight in number)	30%
Weekly Quizzes (six in number)	30%
Midterm Exam	15%
Final Exam	15%
Forum Participation	10%
TOTAL	**100 %**

Participation

Because of the accelerated nature of the course, class participation is very important. Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. If the student doesn't participate during any given week, they will lose the participation points of that week.

Participation means:

1. Present in the forum every week.
2. Regularly checks forum and post all required assignments/discussion questions/ items by the deadlines.
3. Contributes to forum discussions.

Regis University Grading Scale

A	93 – 100	C	73 - 76
A-	90 – 92	C-	70 – 72
B+	87 – 89	D+	67 – 69
B	83 – 86	D	63 – 66
B-	80 – 82	D-	60 – 62
C+	77 – 79	F	59 or Lower

Attendance/Participation

Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

Attendance Policy of the School of Computer and Information Sciences

Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

Late Assignments/Assignment Revisions
Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences
Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, then reduced by 25%. Assignments will not be accepted later than one week after due date.

Incomplete Grade
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

Incomplete Grade Policy of the College of Professional Studies
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

Disability Statement:
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to

classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

Academic Dishonesty

Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.

Non-Public, Confidential or Proprietary Information Compliance

For class projects that relate to a student's employment, the student is responsible for complying with all applicable policies of the employer, including those related to non-public, confidential or proprietary information